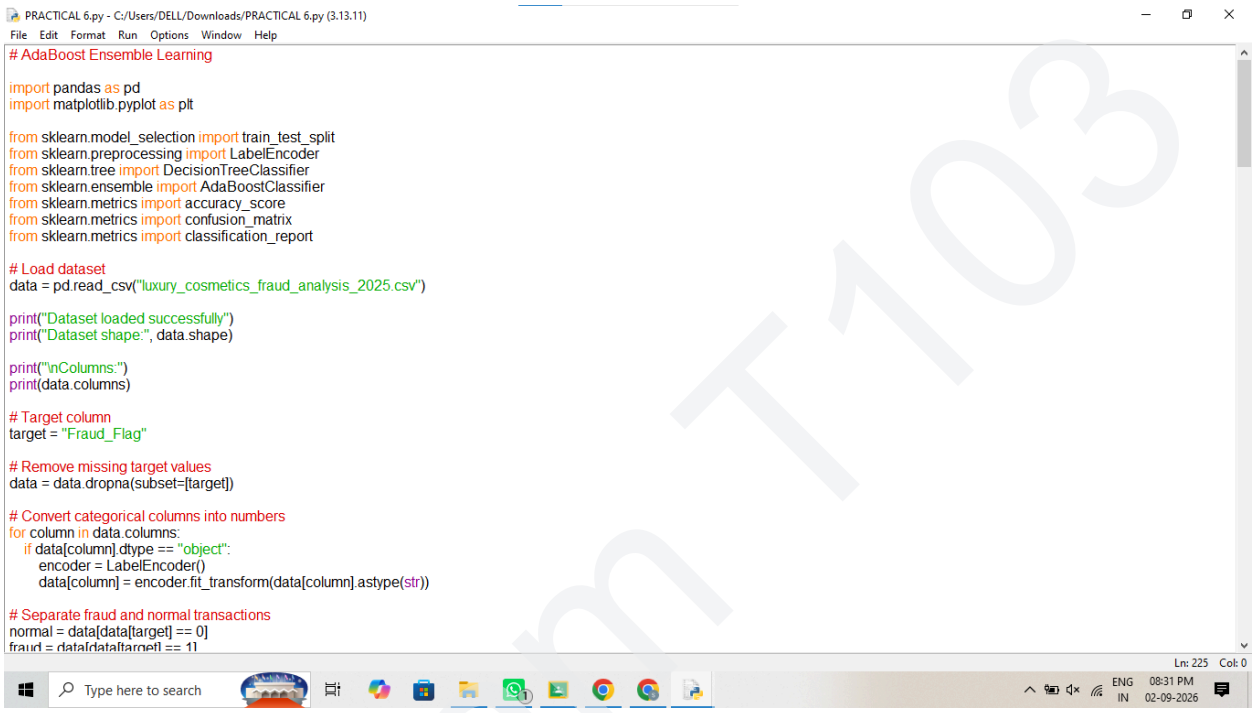
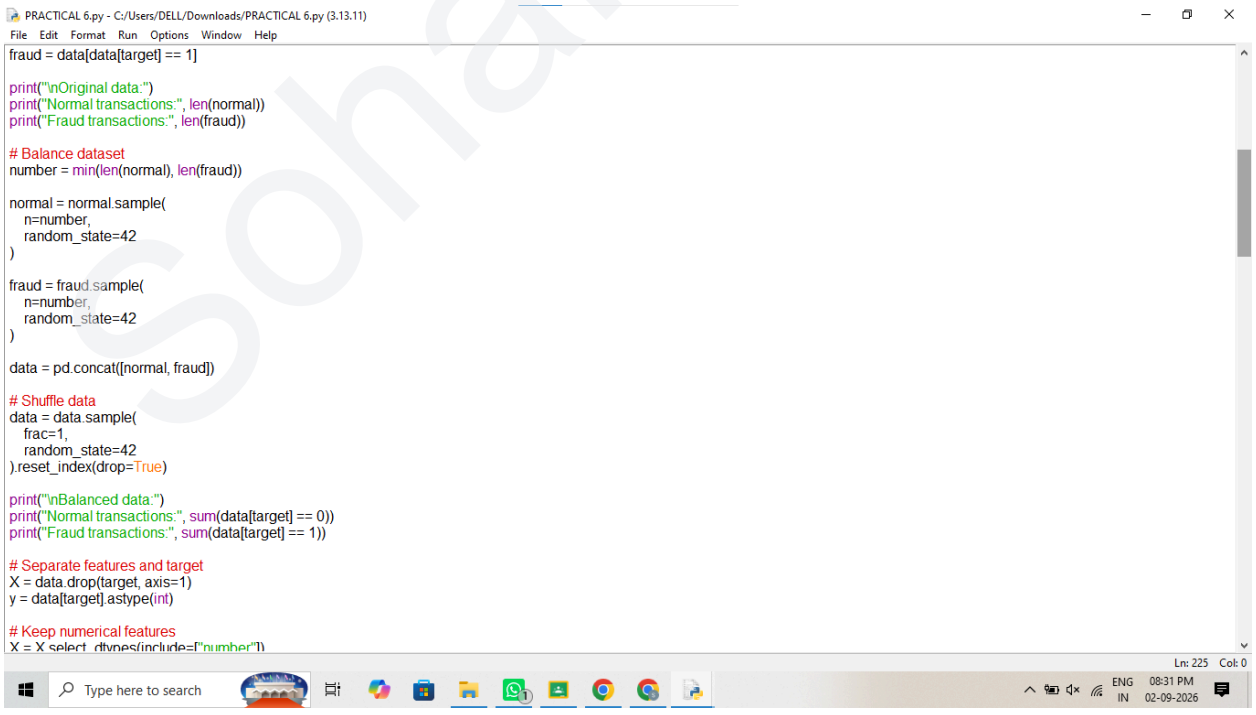
# SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE
# SUBJECT : AI

**Aim: Adaboost Ensemble Learning**
**Implement the Adaboost algorithm to create an ensemble of weak classifiers.**
**Train the ensemble model on a given dataset and evaluate its performance.**
**Compare the results with individual weak classifiers.**

## Input



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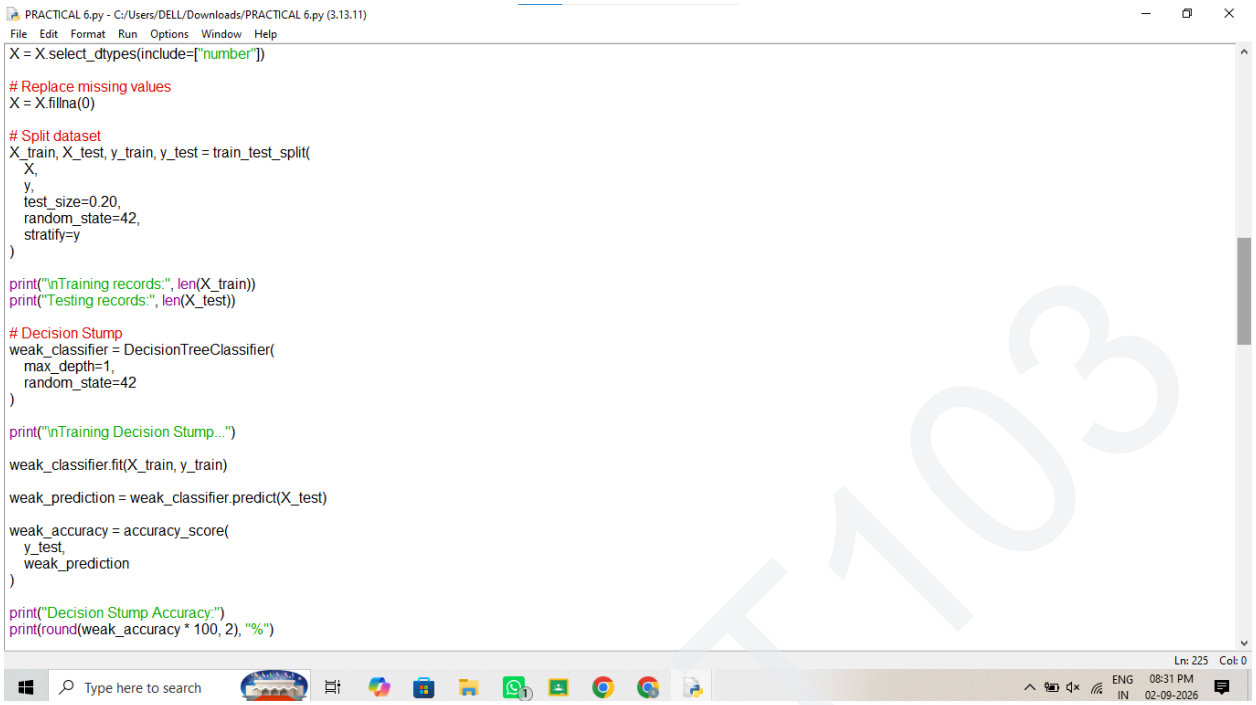


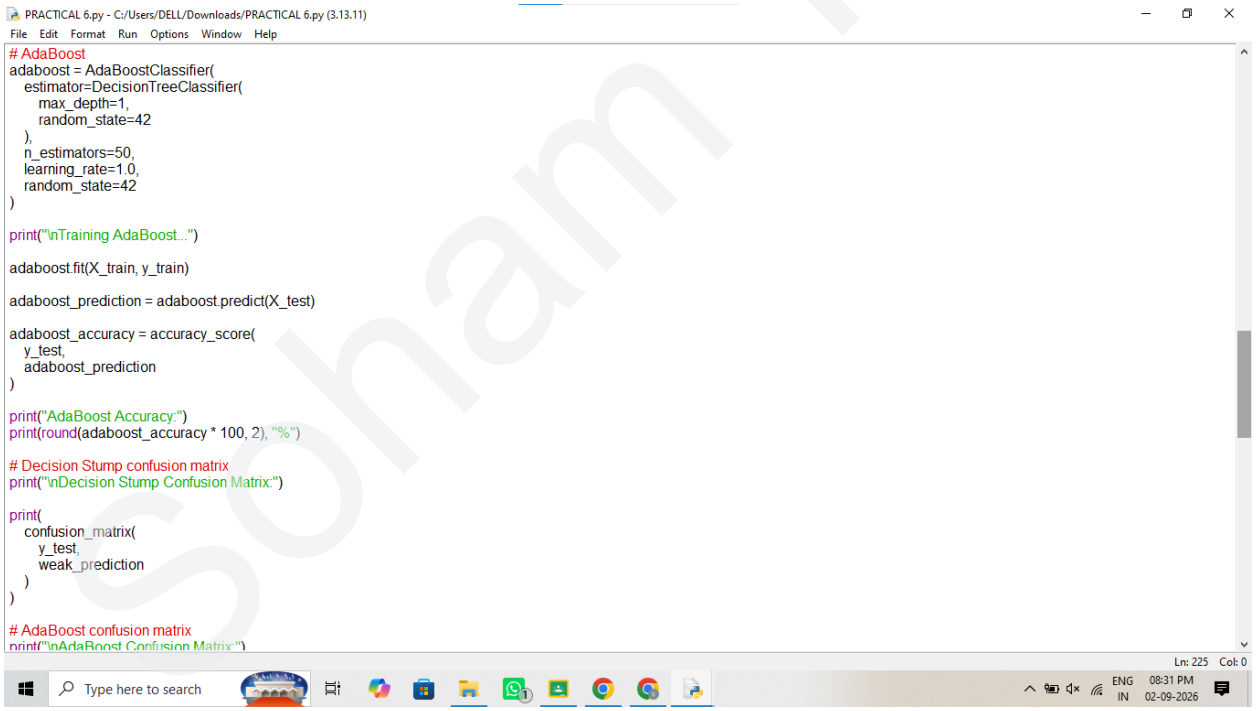




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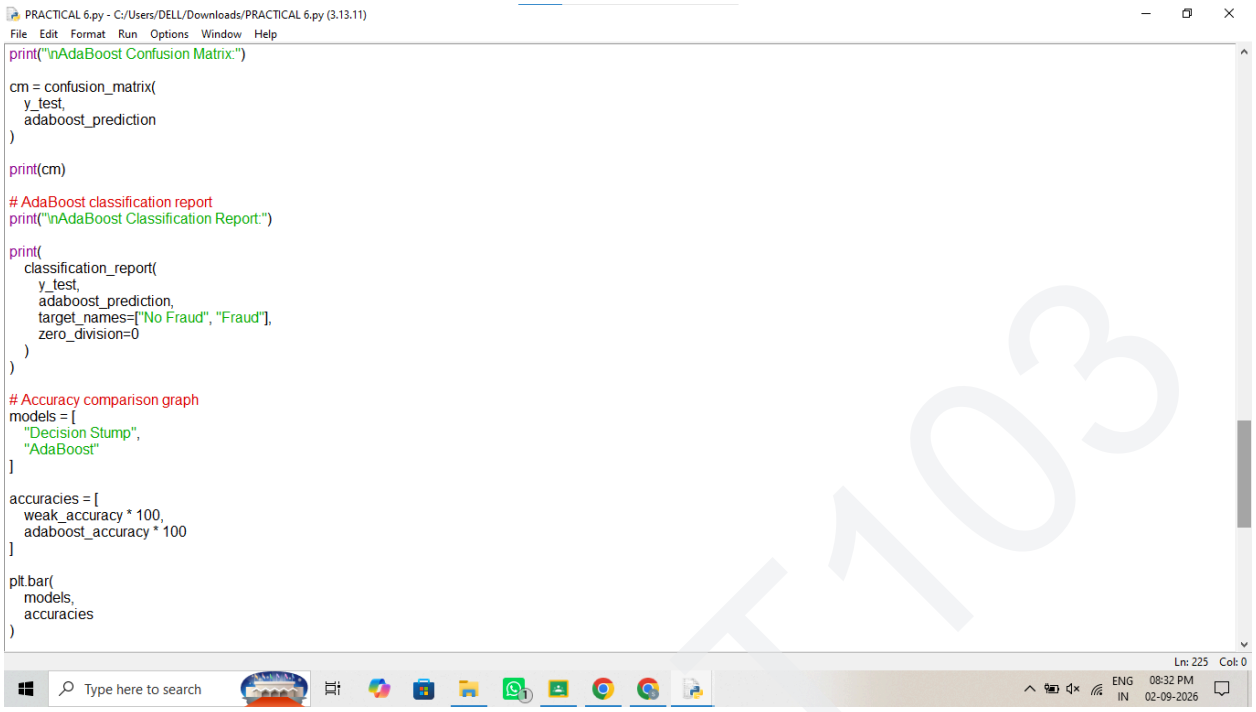


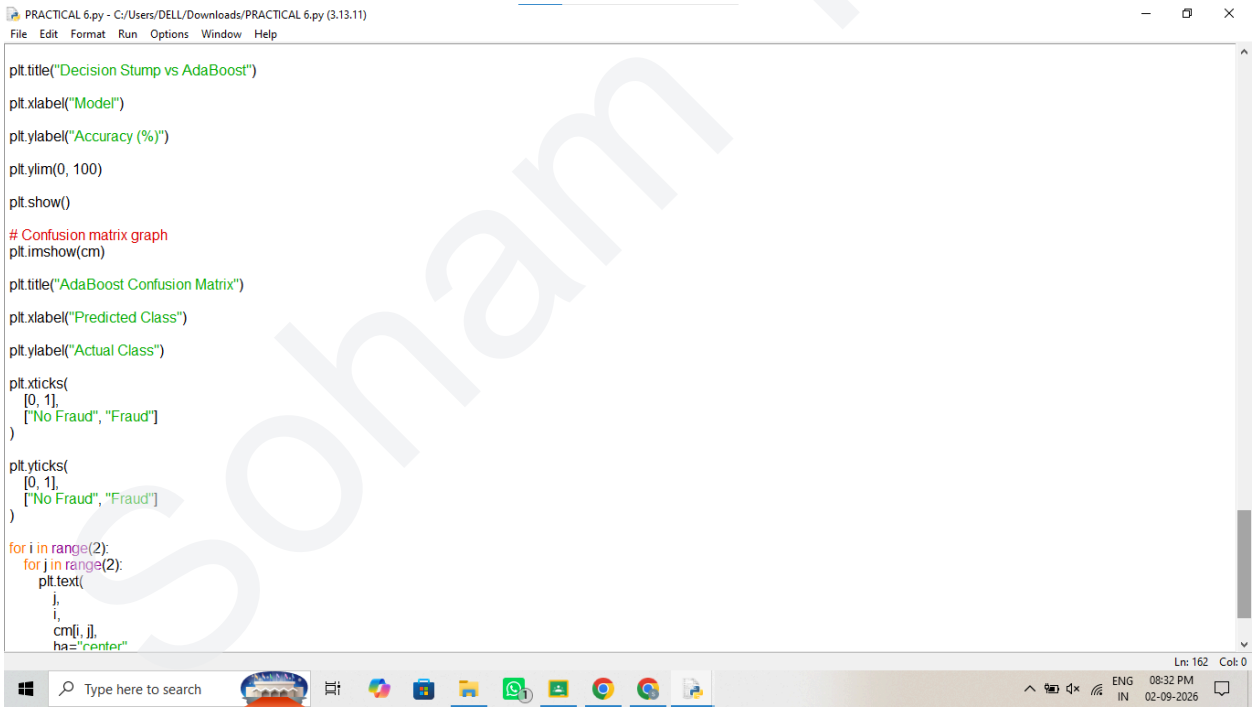




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```
)

plt.yticks(
    [0, 1],
    ["No Fraud", "Fraud"]
)

for i in range(2):
    for j in range(2):
        plt.text(
            j,
            i,
            cm[i, j],
            ha="center",
            va="center"
        )

plt.colorbar()

plt.show()
```

Output

```
IDLE Shell 3.13.11
File  Edit  Shell  Debug  Options  Window  Help
Dataset loaded successfully
Dataset shape: (2133, 16)

Columns:
Index(['Transaction_ID', 'Customer_ID', 'Transaction_Date', 'Transaction_Time',
       'Customer_Age', 'Customer_Loyalty_Tier', 'Location', 'Store_ID',
       'Product_SKU', 'Product_Category', 'Purchase_Amount', 'Payment_Method',
       'Device_Type', 'IP_Address', 'Fraud_Flag', 'Footfall_Count'],
      dtype='object')

Original data:
Normal transactions: 2067
Fraud transactions: 66

Balanced data:
Normal transactions: 66
Fraud transactions: 66

Training records: 105
Testing records: 27

Training Decision Stump...
Decision Stump Accuracy:
44.44 %

Training AdaBoost...
AdaBoost Accuracy:
40.74 %

Decision Stump Confusion Matrix:
[[ 2 12]
 [ 3 10]]

AdaBoost Confusion Matrix:
[[6 8]
 [8 5]]
```

```
AdaBoost Confusion Matrix:
[[6 8]
 [8 5]]

AdaBoost Classification Report:
              precision    recall  f1-score   support

   No Fraud       0.43      0.43      0.43        14
      Fraud       0.38      0.38      0.38        13

   accuracy                           0.41        27
  macro avg       0.41      0.41      0.41        27
weighted avg      0.41      0.41      0.41        27

>>>
```



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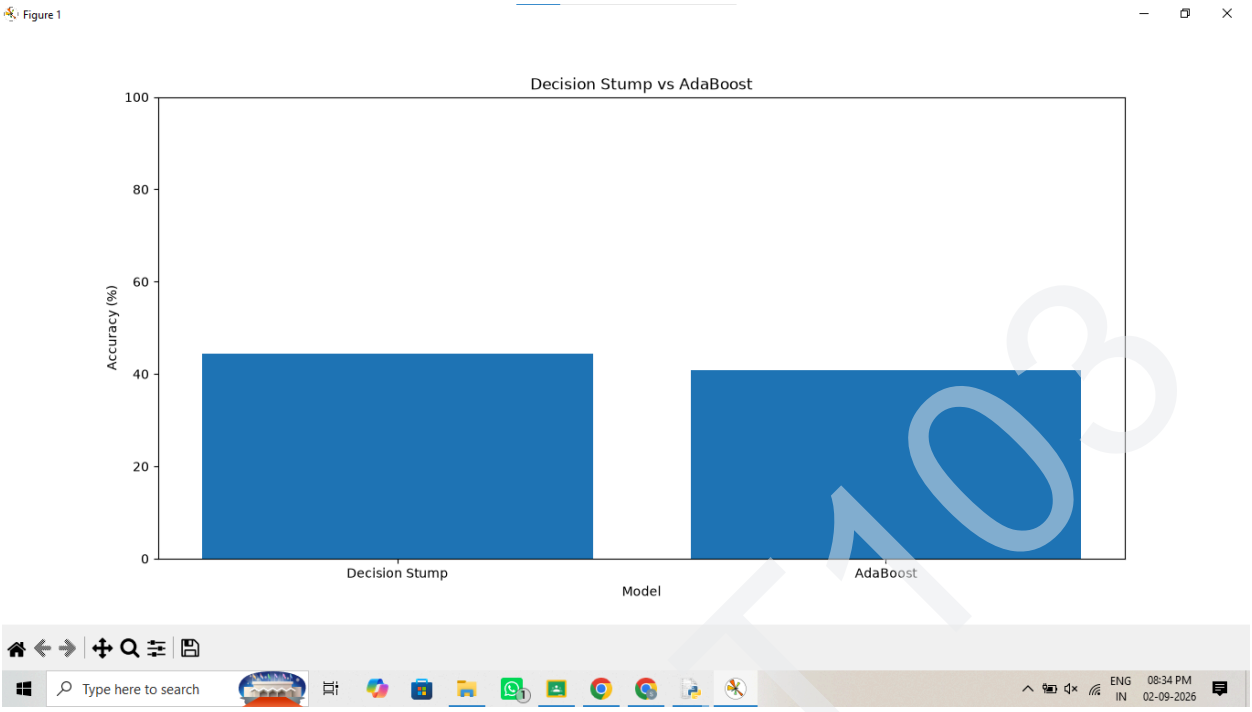


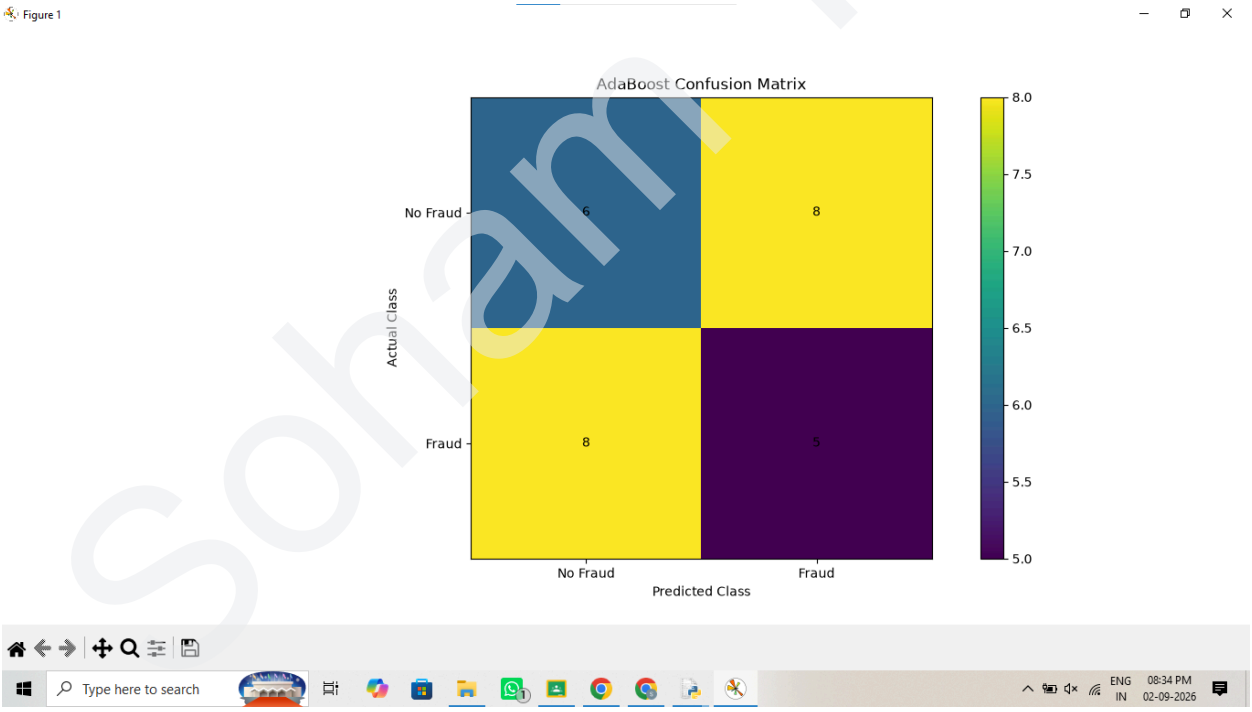




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